

CS 2601 Linear and Convex Optimization

6. Gradient descent (part 1)

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Unconstrained optimization problems

Consider an unconstrained, smooth convex optimization problem

$$\min_{\mathbf{x}} f(\mathbf{x})$$

where f is convex and differentiable on $\mathbb{R}^n$.

The optimal solution satisfies the first-order optimality condition

$$\nabla f(\mathbf{x}^*) = \mathbf{0}$$

In some rare cases, this yields closed-form solutions, e.g.

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

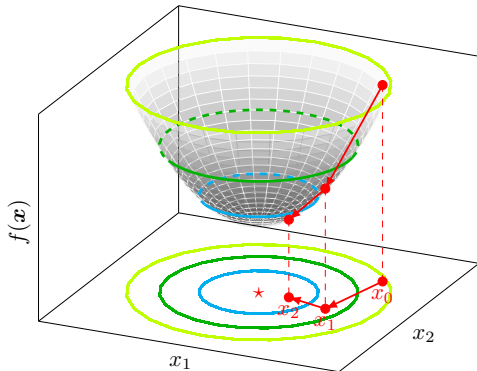
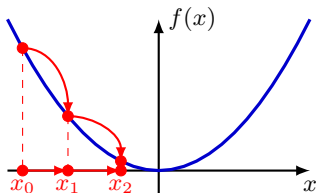
has closed-form solution

$$\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

But in most cases we need numerical algorithms.

Descent method

- 1: choose initial point $\mathbf{x}_0 \in \mathbb{R}^n$
- 2: **repeat**
- 3: choose **descent direction** $\mathbf{d}_k \in \mathbb{R}^n$ and **step size** $t_k > 0$
- 4: $\mathbf{x}_{k+1} = \mathbf{x}_k + t_k \mathbf{d}_k$ s.t. $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$
- 5: **until** stopping criterion is satisfied



Questions

- How to choose $\mathbf{d}_k$ and t_k ?
- Does $\mathbf{x}_k$ converge to $\mathbf{x}^*$?

Descent direction

$\mathbf{d}_k$ is a **descent direction** at $\mathbf{x}_k$ if for all small enough $t > 0$

$$g(t) \triangleq f(\mathbf{x}_k + t\mathbf{d}_k) < f(\mathbf{x}_k) = g(0)$$

For differentiable f (not necessarily convex),

- if $\mathbf{d}_k$ is a descent direction, then $g'(0) = \mathbf{d}_k^T \nabla f(\mathbf{x}_k) \leq 0$;
- if $g'(0) = \mathbf{d}_k^T \nabla f(\mathbf{x}_k) < 0$, then $\mathbf{d}_k$ is a descent direction.

For convex f , by the first-order condition for convexity,

$$f(\mathbf{x}_k) > f(\mathbf{x}_k + t\mathbf{d}_k) \geq f(\mathbf{x}_k) + t\mathbf{d}_k^T \nabla f(\mathbf{x}_k).$$

$\mathbf{d}_k^T \nabla f(\mathbf{x}_k) < 0$ is also **necessary** for $\mathbf{d}_k$ to be a descent direction.

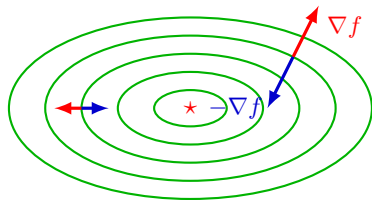
For convex differentiable f ,

$$\mathbf{d}_k \text{ is a descent direction} \iff \mathbf{d}_k^T \nabla f(\mathbf{x}_k) < 0$$

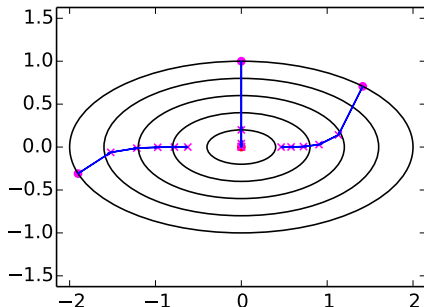
Gradient descent

Choose $\mathbf{d}_k = -\nabla f(\mathbf{x}_k)$, $\mathbf{d}_k^T \nabla f(\mathbf{x}_k) = -\|\nabla f(\mathbf{x}_k)\|_2^2 < 0$ unless $\nabla f(\mathbf{x}_k) = \mathbf{0}$.

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t_k \nabla f(\mathbf{x}_k)$$



level curves of $f(x_1, x_2) = \frac{x_1^2}{4} + x_2^2$



Question. What happens if $\nabla f(\mathbf{x}_k) = \mathbf{0}$?

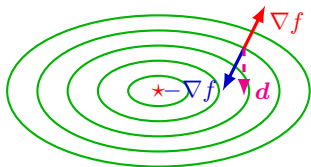
Max-rate descending direction

$-\nabla f(\mathbf{x}_k)$ is the direction of fastest rate of decrease of f at $\mathbf{x}_k$

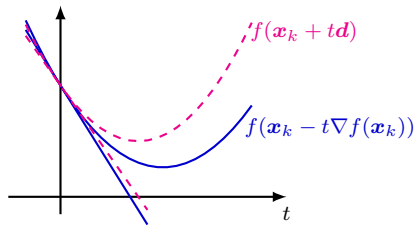
- If $\|\mathbf{d}_k\|_2 = 1$,

$$\lim_{t \downarrow 0} \frac{f(\mathbf{x}_k) - f(\mathbf{x}_k + t\mathbf{d}_k)}{t} = -\mathbf{d}_k^T \nabla f(\mathbf{x}_k) \leq \|\nabla f(\mathbf{x}_k)\|_2$$

with equality iff $\mathbf{d}_k = -\nabla f(\mathbf{x}_k) / \|\nabla f(\mathbf{x}_k)\|_2$



level curves of $f(x_1, x_2) = \frac{x_1^2}{4} + x_2^2$



Gradient descent algorithm

- 1: initialization $\mathbf{x} \leftarrow \mathbf{x}_0 \in \mathbb{R}^n$
- 2: **while** $\|\nabla f(\mathbf{x})\| > \delta$ **do**
- 3: $\mathbf{x} \leftarrow \mathbf{x} - t \nabla f(\mathbf{x})$
- 4: **end while**
- 5: **return** $\mathbf{x}$

Step size (aka **learning rate** in machine learning)

- the above algorithm uses constant step size t for all iterations
- there are other methods for choosing t for each iteration, e.g. **exact line search**, **backtracking line search**

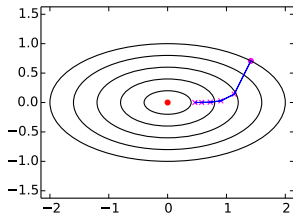
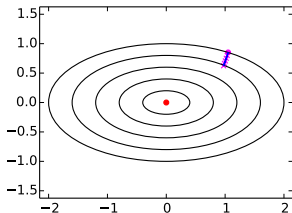
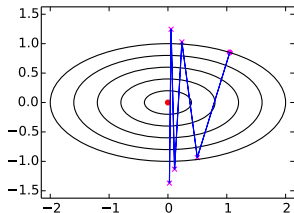
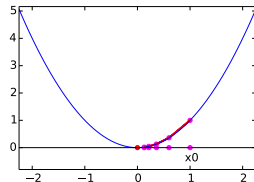
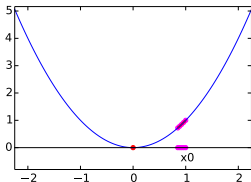
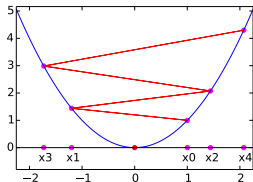
Stopping criterion

- ideally, stop if $\nabla f(\mathbf{x}) = 0$ (optimality condition), but impractical
- more practical: stop when $\|\nabla f(\mathbf{x})\| \leq \delta$ for some small δ
- other criteria: $|f(\mathbf{x}_{\text{new}}) - f(\mathbf{x}_{\text{old}})| \leq \delta$, $\frac{|f(\mathbf{x}_{\text{new}}) - f(\mathbf{x}_{\text{old}})|}{|f(\mathbf{x}_{\text{old}})|} \leq \delta$, ...
- in practice, also stop if maximum # of iterations is reached

Large vs. small step size

Consider constant step size. How large should the step size be?

- Too large: may oscillate and diverge
- Too small: may be too slow
- “Just right”: fast convergence



1D example

Consider $f(x) = \frac{1}{2}ax^2$, where $a > 0$.

- gradient step

$$x_{k+1} = x_k - tf'(x_k) = (1 - at)x_k$$

- descent condition

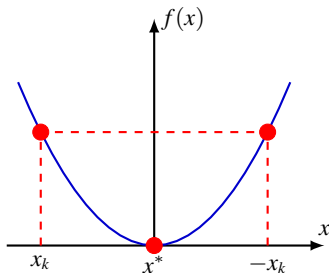
$$f(x_{k+1}) < f(x_k) \iff |1 - at| < 1 \iff 0 < t < \frac{2}{a}$$

- $x_k = (1 - at)^k x_0 \rightarrow x^* = 0$ geometrically for such t

Note f satisfies

- $|f'(x) - f'(y)| = a|x - y|$
- $f''(x) = a$

f' is so-called **Lipschitz continuous**
and t is roughly the order of $\frac{1}{a}$.



Lipschitz continuity

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **Lipschitz continuous** with **Lipschitz constant** $L > 0$, or simply **L -Lipschitz**, if

$$\|f(\mathbf{x}) - f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y}$$

Note. Lipschitz continuity can be defined with respect to any norms. But we will assume the norms in the above definition are the 2-norms in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively, unless stated otherwise.

Note. Lipschitz continuity implies uniform continuity.

Example. $f(x) = ax$ is $|a|$ -Lipschitz, $|f(x) - f(y)| = |a| \cdot |x - y|$

Example. $f(x) = |x|$ is 1-Lipschitz, $|f(x) - f(y)| = ||x| - |y|| \leq |x - y|$

Example. $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$ is $\|\mathbf{a}\|$ -Lipschitz, $|\mathbf{a}^T \mathbf{x} - \mathbf{a}^T \mathbf{y}| \leq \|\mathbf{a}\| \cdot \|\mathbf{x} - \mathbf{y}\|$ by the Cauchy-Schwarz inequality.

Lipschitz continuity (cont'd)

Example. Let $\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$. $f(\mathbf{x}) = \mathbf{Q}\mathbf{x} = (x_1, 2x_2)$ is 2-Lipschitz.

$$f(\mathbf{x}) - f(\mathbf{y}) = (x_1 - y_1, 2x_2 - 2y_2) = (d_1, 2d_2)$$

$$\|f(\mathbf{x}) - f(\mathbf{y})\| = \sqrt{d_1^2 + 4d_2^2} \leq 2\sqrt{d_1^2 + d_2^2} = 2\|\mathbf{x} - \mathbf{y}\|$$

More generally, $f(\mathbf{x}) = \mathbf{Q}\mathbf{x}$ with $\mathbf{Q} \succeq \mathbf{0}$ is $\lambda_{\max}(\mathbf{Q})$ -Lipschitz, where $\lambda_{\max}(\mathbf{Q})$ is the largest eigenvalue of $\mathbf{Q}$ ¹.

Proof. Let $\mathbf{d} = \mathbf{x} - \mathbf{y}$. By slide 32 of §2,

$$\|f(\mathbf{x}) - f(\mathbf{y})\| = \|\mathbf{Q}\mathbf{d}\| = \sqrt{\mathbf{d}^T \mathbf{Q}^2 \mathbf{d}} \leq \sqrt{\lambda_{\max}(\mathbf{Q}^2) \|\mathbf{d}\|^2} = \lambda_{\max}(\mathbf{Q}) \|\mathbf{x} - \mathbf{y}\|$$

The last equality uses the fact $\lambda_{\max}(\mathbf{Q}^2) = \lambda_{\max}^2(\mathbf{Q})$.

¹For general $\mathbf{Q}$, $f(\mathbf{x}) = \mathbf{Q}\mathbf{x}$ is $\sigma_{\max}(\mathbf{Q})$ -Lipschitz, where $\sigma_{\max}(\mathbf{Q}) = \sqrt{\lambda_{\max}(\mathbf{Q}^T \mathbf{Q})}$ is the largest singular value of $\mathbf{Q}$.

L -smoothness

A function is L -smooth if it is differentiable and its gradient is L -Lipschitz, i.e.

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y}$$

Note. L upper bounds the rate of change of ∇f

Example. $f(x) = \frac{1}{2}ax^2$ is $|a|$ -smooth, since $f'(x) = ax$ is $|a|$ -Lipschitz

Example. $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{Q} \mathbf{x}$ with $\mathbf{Q} \succeq \mathbf{O}$ is $\lambda_{\max}(\mathbf{Q})$ -smooth, since $\nabla f(\mathbf{x}) = \mathbf{Q} \mathbf{x}$ is $\lambda_{\max}(\mathbf{Q})$ -Lipschitz.

With $\mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, we obtain $f(\mathbf{x}) = \frac{1}{2}x_1^2 + x_2^2$ is 2-smooth.

Lemma. A twice continuously differentiable **convex** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth iff $\nabla^2 f(\mathbf{x}) \preceq L\mathbf{I}$, meaning $L\mathbf{I} - \nabla^2 f(\mathbf{x}) \succeq \mathbf{O}$, or equivalently $\lambda_{\max}(\nabla^2 f(\mathbf{x})) \leq L$.

Appendix: Second-order condition for L -smoothness

Lemma. A twice continuously differentiable $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth iff for any $\mathbf{x}$, $-L\mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq L\mathbf{I}$, or equivalently $|\lambda| \leq L$ for all eigenvalues λ of $\nabla^2 f(\mathbf{x})$.

Proof. “ $\Leftarrow$ ”. Assume $-L\mathbf{I} \preceq \nabla^2 f(\mathbf{x}) \preceq L\mathbf{I}$ for all $\mathbf{x}$. Let $g(\mathbf{x}) = \mathbf{u}^T \nabla f(\mathbf{x})$. Then $\nabla g(\mathbf{x}) = \nabla^2 f(\mathbf{x})\mathbf{u}$.

- By slide 32 of §2, $\|\nabla g(\mathbf{x})\| \leq \lambda_{\max}(\nabla^2 f(\mathbf{x}))\|\mathbf{u}\| \leq L\|\mathbf{u}\|$.
- By the Mean Value Theorem,

$$\mathbf{u}^T [\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})] = g(\mathbf{x}) - g(\mathbf{y}) = \nabla g(\mathbf{z})^T (\mathbf{x} - \mathbf{y}) \leq L\|\mathbf{u}\| \cdot \|\mathbf{x} - \mathbf{y}\|$$

Setting $\mathbf{u} = \nabla f(\mathbf{x}) - \nabla f(\mathbf{y})$ yields $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$.

“ $\Rightarrow$ ”. Assume f is L -smooth. Let $\mathbf{d}$ be an eigenvector of $\nabla^2 f(\mathbf{x})$ with associated eigenvalue λ . By L -smoothness,

$$\|\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x})\| \leq L\|t\mathbf{d}\| = tL\|\mathbf{d}\|$$

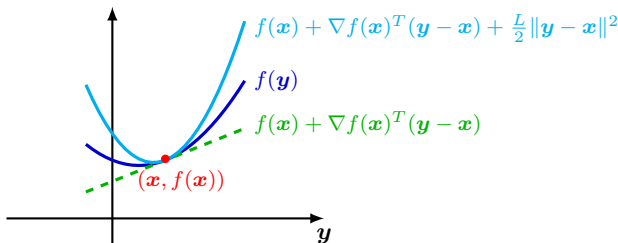
Dividing both sides by t and letting $t \rightarrow 0$,

$$|\lambda| \cdot \|\mathbf{d}\| = \|\nabla^2 f(\mathbf{x})\mathbf{d}\| \leq L\|\mathbf{d}\| \implies |\lambda| \leq L$$

Quadratic upper bound

Lemma. If f is L -smooth, then

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2$$



Note. The upper bound does **not** assume the convexity of f .

If $\nabla^2 f(\mathbf{x}) \preceq L\mathbf{I}$, the upper bound is intuitive from the Taylor expansion

$$f(\mathbf{y}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x}) + \frac{1}{2} (\mathbf{y} - \mathbf{x})^T \nabla^2 f(\mathbf{x} + \theta(\mathbf{y} - \mathbf{x})) (\mathbf{y} - \mathbf{x})$$

for some $\theta \in [0, 1]$. (Check $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{Q} \mathbf{x}$)

Proof

First prove the 1D case. Let $g(t)$ be L_g -smooth, $|g'(t) - g'(s)| \leq L_g|t - s|$.

$$\begin{aligned} g(1) &= g(0) + \int_0^1 g'(t) dt \\ &= g(0) + g'(0) + \int_0^1 [g'(t) - g'(0)] dt \\ &\leq g(0) + g'(0) + \int_0^1 L_g t dt \quad \text{since } |g'(t) - g'(0)| \leq L_g t \\ &= g(0) + g'(0) + \frac{1}{2} L_g \end{aligned}$$

For the general case, apply the above to $g(t) = f(\mathbf{x} + t\mathbf{d})$ with $\mathbf{d} = \mathbf{y} - \mathbf{x}$ and $L_g = L\|\mathbf{d}\|^2$. To check the smoothness of g ,

$$\begin{aligned} |g'(t) - g'(s)| &= |[\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x} + s\mathbf{d})]^T \mathbf{d}| \\ &\leq \|\nabla f(\mathbf{x} + t\mathbf{d}) - \nabla f(\mathbf{x} + s\mathbf{d})\| \cdot \|\mathbf{d}\| \quad \text{Cauchy-Schwarz} \\ &\leq (t - s)L\|\mathbf{d}\|^2 \quad \text{\textit{f is L-smooth}} \end{aligned}$$

Consequence of quadratic upper bound

For L -smooth f , the sequence $\{\mathbf{x}_k\}$ produced by gradient descent satisfies

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - t \left(1 - \frac{Lt}{2}\right) \|\nabla f(\mathbf{x}_k)\|^2$$

Proof. Plugging in $\mathbf{x} = \mathbf{x}_k$ and $\mathbf{y} = \mathbf{x}_{k+1} = \mathbf{x}_k - t\nabla f(\mathbf{x}_k)$ in the quadratic upper bound,

$$\begin{aligned} f(\mathbf{x}_{k+1}) &\leq f(\mathbf{x}_k) - t\|\nabla f(\mathbf{x}_k)\|^2 + \frac{L}{2}t^2\|\nabla f(\mathbf{x}_k)\|^2 \\ &= f(\mathbf{x}_k) - t \left(1 - \frac{Lt}{2}\right) \|\nabla f(\mathbf{x}_k)\|^2 \end{aligned}$$

Note. If $\nabla f(\mathbf{x}_k) \neq 0$ and $0 < t < \frac{2}{L}$, then $f(\mathbf{x}_{k+1}) < f(\mathbf{x}_k)$, so gradient descent with step size $t \in (0, 2/L)$ is indeed a descent method.

Note. We can lower bound the decrease in function value in each step. In particular, for $0 < t \leq \frac{1}{L}$,

$$f(\mathbf{x}_k) - f(\mathbf{x}_{k+1}) \geq \frac{t}{2} \|\nabla f(\mathbf{x}_k)\|^2$$

Convergence analysis

Theorem. If f is convex and L -smooth, and $\mathbf{x}^*$ is a minimum of f , then for step size $t \in (0, \frac{1}{L}]$, the sequence $\{\mathbf{x}_k\}$ produced by the gradient descent algorithm satisfies

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2tk}$$

Notes.

- $f(\mathbf{x}_k) \downarrow f^*$ as $k \rightarrow \infty$.
- Any limiting point of $\mathbf{x}_k$ is an optimal solution.
- The rate of convergence is $O(1/k)$, i.e. # of iterations to guarantee $f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \epsilon$ is $O(1/\epsilon)$. For $\epsilon = 10^{-p}$, $k = O(10^p)$, exponential in the number of significant digits!
- Faster convergence with larger t ; best $t = \frac{1}{L}$, but L is unknown.
- Good initial guess helps.

Gradient flow

Consider a **gradient flow**, a continuous version of gradient descent,

$$\frac{d}{ds}\tilde{\mathbf{x}}_s = -\nabla f(\tilde{\mathbf{x}}_s)$$

$f(\tilde{\mathbf{x}}_s)$ is nonincreasing,

$$\frac{d}{ds}f(\tilde{\mathbf{x}}_s) = \langle \nabla f(\tilde{\mathbf{x}}_s), \frac{d}{ds}\tilde{\mathbf{x}}_s \rangle = -\|\nabla f(\tilde{\mathbf{x}}_s)\|^2 \leq 0$$

If f is convex and $\mathbf{x}^*$ is a minimum of f ,

$$\frac{d}{ds}\|\tilde{\mathbf{x}}_s - \mathbf{x}^*\|^2 = 2\langle \tilde{\mathbf{x}}_s - \mathbf{x}^*, \frac{d}{ds}\tilde{\mathbf{x}}_s \rangle = 2\langle \nabla f(\tilde{\mathbf{x}}_s), \mathbf{x}^* - \tilde{\mathbf{x}}_s \rangle \leq 2[f(\mathbf{x}^*) - f(\tilde{\mathbf{x}}_s)]$$

Integrating from 0 to T and using $f(\tilde{\mathbf{x}}_s) \geq f(\tilde{\mathbf{x}}_T)$,

$$\|\tilde{\mathbf{x}}_T - \mathbf{x}^*\|^2 - \|\tilde{\mathbf{x}}_0 - \mathbf{x}^*\|^2 \leq \int_0^T 2[f(\mathbf{x}^*) - f(\tilde{\mathbf{x}}_s)]ds \leq 2T[f(\mathbf{x}^*) - f(\tilde{\mathbf{x}}_T)]$$

so

$$f(\tilde{\mathbf{x}}_T) - f(\mathbf{x}^*) \leq \frac{\|\tilde{\mathbf{x}}_0 - \mathbf{x}^*\|^2}{2T}$$

Proof

Discretizing the gradient flow by Euler's method with step size t ,

$$\frac{\tilde{\mathbf{x}}_{(k+1)t} - \tilde{\mathbf{x}}_{kt}}{t} \approx -\nabla f(\tilde{\mathbf{x}}_{kt})$$

1. By the basic gradient step $\mathbf{x}_{k+1} = \mathbf{x}_k - t\nabla f(\mathbf{x}_k)$,

$$\begin{aligned}\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 - \|\mathbf{x}_k - \mathbf{x}^*\|^2 &= \|\mathbf{x}_k - t\nabla f(\mathbf{x}_k) - \mathbf{x}^*\|^2 - \|\mathbf{x}_k - \mathbf{x}^*\|^2 \\ &= 2t\nabla f(\mathbf{x}_k)^T(\mathbf{x}^* - \mathbf{x}_k) + t^2\|\nabla f(\mathbf{x}_k)\|^2\end{aligned}$$

2. By the first-order condition for convexity, the first term is upper bounded by

$$2t\nabla f(\mathbf{x}_k)^T(\mathbf{x}^* - \mathbf{x}_k) \leq 2t[f(\mathbf{x}^*) - f(\mathbf{x}_k)]$$

3. By L -smoothness (the last inequality on slide 15), the second (extra!) term is upper bounded by

$$t^2\|\nabla f(\mathbf{x}_k)\|^2 \leq 2t[f(\mathbf{x}_k) - f(\mathbf{x}_{k+1})]$$

Proof (cont'd)

4. Plugging 2 and 3 into 1,

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 - \|\mathbf{x}_k - \mathbf{x}^*\|^2 \leq 2t[f(\mathbf{x}^*) - f(\mathbf{x}_{k+1})]$$

5. Summing over k from 0 to $N - 1$,

$$\|\mathbf{x}_N - \mathbf{x}^*\|^2 - \|\mathbf{x}_0 - \mathbf{x}^*\|^2 \leq 2t \sum_{k=0}^{N-1} [f(\mathbf{x}^*) - f(\mathbf{x}_{k+1})]$$

6. By the descent property $f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k)$,

$$\|\mathbf{x}_N - \mathbf{x}^*\|^2 - \|\mathbf{x}_0 - \mathbf{x}^*\|^2 \leq 2tN[f(\mathbf{x}^*) - f(\mathbf{x}_N)]$$

so

$$f(\mathbf{x}_N) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|^2}{2tN}$$